

MISSOURI S&T
Department of Mathematics & Statistics

Stat 3113 - Section 102
Exam 3

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FALL 2024

Student Name: _____ *Key!*

Good luck and happy reading work!

Instructions:

1. Be sure to print your first and last name in the spaces provided.
 2. Work all problems. **Show all work.** Full credit will be given only if work is shown which *fully justified* your answer.
 3. There should be sufficient space under each problem in which to show your work. **Do not use the back of the page where the problem is on.**
 4. There are **6 problems**. Point values are indicated (in parentheses) for each problem. The total number of points is 100.
 5. Turn off your cell phone.
 6. Calculator is allowed in this Exam.
 7. There are 10 pages in this Exam 3 set (including this page). **Please check to make sure you have 10 pages with 6 problems.**
 8. Please ask only your instructor if you have any questions during the exam.
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Distribution of Marks

Question	Points	Score
1	10	
2	12	
3	10	
4	18	
5	15	
6	35	
Total:	100	

1. (10 points) *For the multiple choice questions, circle the correct answer.*

(a) (2 points) In a One-Way ANOVA test, if the p-value is less than 0.05, what conclusion can you draw?

- ☐ There is no evidence to suggest that the group means are different.
- ☒ At least one group mean is significantly different from the others.
- ☐ All group means are significantly different from each other.
- ☐ The test is invalid because the p-value is too low.

(b) (2 points) Which of the following measures indicates how well the regression line fits the data?

- ☐ The slope of the regression line.
- ☒ The R^2 (coefficient of determination).
- ☐ The residuals.
- ☐ The correlation coefficient.

(c) (2 points) In a Two-Way ANOVA, what does an interaction effect represent?

- ☐ The effect of one factor on the dependent variable, ignoring the other factor.
- ☒ The combined effect of two factors on the response variable.
- ☐ The difference in variances within each group.
- ☐ The correlation between the two factors.

(d) (2 points) Which of the following patterns in a scatter plot suggests a positive linear relationship between two variables?

- ☐ Points scattered randomly without any trend.
- ☐ Points forming a downward-sloping pattern from left to right.
- ☒ Points forming an upward-sloping pattern from left to right.
- ☐ Points clustering tightly around a horizontal line.

(e) (2 points) What is the possible range of Pearson's correlation coefficient (r)?

- ☐ 0 to 1
- ☐ $-\infty$ to $+\infty$
- ☒ -1 to +1
- ☐ 0 to $+\infty$

2. (12 points) **True/False** Indicate whether the following statements are true or false.

- (a) ☒ True ☐ False In a One-Way ANOVA, an individual confidence level (α_{cw}) is the success rate of a procedure for constructing a single confidence interval (or conducting a single hypothesis test).
- (b) ☒ True ☐ False In a One-Way ANOVA, the true errors are unknown, but our estimates of them are called residuals
- (c) ☒ True ☐ False In a Two-Way ANOVA, when the number of replicates is the same for each treatment, the design is said to be balanced.
- (d) ☐ True ☒ False A Pearson's correlation coefficient of zero means NO relationship.
- (e) ☒ True ☐ False Every influential point is an outlier, but not every outlier is an influential point
- (f) ☐ True ☒ False Causation can always be inferred from a high correlation between two quantitative variables.

3. (10 points) **MATCHING:** For problems (a)-(e), write the letter from the list below that best matches the description. Note: Each answer may be used once, more than once, or not at all.

K Line plot

M The slope β_1

B Nuisance Variable

D Interaction effect

P Least squares

Q Main effect

R Causal effect

S Coefficient of determination

T Residual plot

L Lurking Variable

- (a) (2 points) K A scatter plot with a regression line that best fits the data points.
- (b) (2 points) P The method used to find the optimal line that best describes the relationship in linear regression.
- (c) (2 points) Q The change in the response resulting from modifying the level of a factor, while holding other factors constant in a factorial experiment.
- (d) (2 points) D It measures the degree to which two or more factors interact and influence the response.
- (e) (2 points) L A variable not included in the study that has a significant impact on the relationship between the response and the explanatory variable.

4. (18 points) An experimental study is conducted to determine whether the average blood flow rate differs when an artery collapses at different levels of stenosis (narrowing of the artery). Participants are randomly assigned to one of several groups representing different levels of stenosis. The blood flow rate is measured under controlled conditions when the artery collapses. The goal is to assess whether the average flow rate differs significantly across the levels of stenosis.

Data was analyzed using JMP.

Use $\alpha = 0.05$ to answer the questions that follow.

- (a) (3 points) In the study on blood flow rate and stenosis, which are the factors and response variable?
- ☐ Factors: Blood flow rate, Response variable: Stenosis level
 - ☒ Factors: Stenosis level, Response variable: Blood flow rate
 - ☐ Factors: Age, Response variable: Stenosis level
 - ☐ Factors: Blood flow rate, Age, Response variable: Stenosis level
- (b) (3 points) Which of the following represents the null and alternative hypotheses for comparing the group means?
- ☒ $H_0: \mu_1 = \mu_2 = \mu_3, H_a: \text{at least one } \mu \text{ differs}$
 - ☐ $H_0: \mu_1 \neq \mu_2 \neq \mu_3, H_a: \text{all } \mu\text{'s are equal}$
- (c) (4 points) Given the following ANOVA table in Figure 1 what conclusion should be drawn based on the p-value?

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Stenosis	2	204.02019	102.010	23.5747	<.0001*
Error	32	138.46723	4.327		
C. Total	34	342.48743			

Figure 1: ANOVA Table for blood flow rate at different levels of stenosis.

- ☐ Fail to reject the null hypothesis and conclude that the average blood flow rate is the same across the stenosis levels.
 - ☒ Reject the null hypothesis and conclude that at least one stenosis level has a significantly different blood flow rate.
- (d) (2 points) Based on the ANOVA results, should you proceed with a post-ANOVA multiple comparison test to identify which stenosis levels differ significantly from each other?
- ☒ Yes, since the ANOVA shows significant differences between the groups.
 - ☐ No, because the ANOVA result already tells us which groups differ.

- (e) (2 points) Based on the connecting letter and ordered differences reports for blood flow rate across stenosis levels (1, 2, 3), which conclusion is correct?

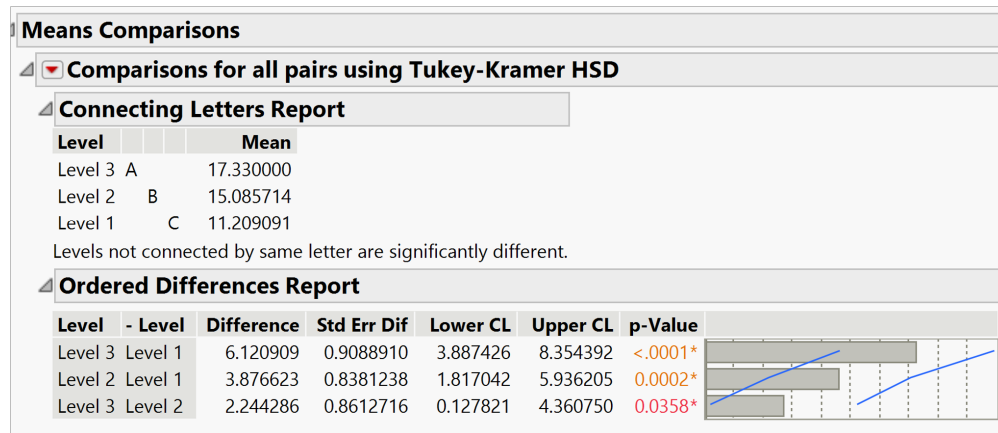


Figure 2: Post-ANOVA multiple comparison results for blood flow rate across stenosis levels.

- ☒ All levels (1, 2, 3) are statistically different from each other.
- ☐ There are no significant differences between any of the levels.
- ☐ Level 1 and Level 3 are not significantly different, but both differ from Level 2.
- ☐ Level 1 and Level 2 are not significantly different, but both are different from Level 3.

- (f) (4 points) Based on the diagnostic plots (Normal Probability Plot and Residual vs. Predicted Plot), which of the following conclusions can be drawn?

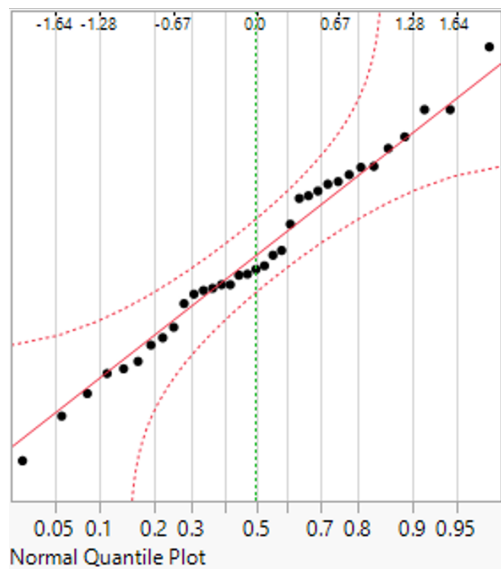


Figure 3: Normal Probability Plot

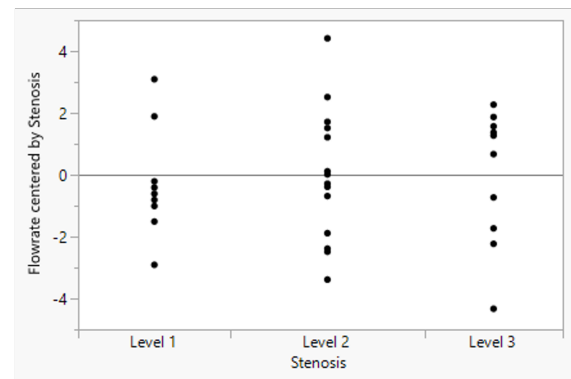


Figure 4: Residual vs. Predicted Plot

Figure 5: Diagnostic Plots for checking ANOVA Assumptions.

- ☒ The model appears to meet the assumptions of normality and constant variance.
- ☐ The Normal Probability Plot shows significant deviations from normality, indicating a need for model adjustment.
- ☐ The Residual vs. Predicted Plot shows a clear non-random pattern, suggesting a violation of the constant variance assumption.
- ☐ The plots indicate the presence of outliers that need to be removed before further analysis.

5. (15 points) A factorial experiment was conducted to estimate the effects of three factors on the yield of a corn. The factors and their levels were:

A: Amount of fertilizer (low or high)

B: Sun light (low or high)

C: Type of Soil (rich or not rich)

Three replicates were obtained for each treatment combination. Data was analyzed using JMP and the following outputs obtained.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	7	192.30018	27.47145	3.5030
Error	16	125.47580	7.8422	Prob > F
C. Total	23	317.77598		0.0179*

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Amount of fertilizer					0.0155*
Sun light	1	1	44.717400	5.7021	0.0296*
Type of soil	1	1	5.226667	0.6665	0.4263
Amount of fertilizer*Sun light	1	1	60.483750	7.7126	0.0135*
Amount of fertilizer*Type of soil	1	1	10.746817	1.3704	0.2589
Sun light*Type of soil	1	1	6.762817	0.8624	0.3669
Amount of fertilizer*Sun light*Type of soil	1	1	6.826667	0.8705	0.3647

Use $\alpha = 0.05$ to answer the questions that follow. You can assume all ANOVA assumptions are satisfied

- (a) (6 points) Calculate the appropriate test statistic needed to ascertain the overall significance or adequacy of the ANOVA model.

$$F\text{-Ratio} = \frac{MSTr}{MSE} = \frac{27.47145}{7.8422} = 3.5030$$

$$\text{with } MSTr = \frac{SSTr}{df} = \frac{192.30018}{7} = 27.47145$$

$$F\text{-Ratio} = \boxed{3.5030}$$

- (b) (3 points) Based on the p-value from the ANOVA table:

☐ True ☒ False The ANOVA model is NOT adequate for the data since the p-value is less than the significance level.

- (c) (3 points) Is a three-way interaction present or significant?

☐ Yes, the three-way interaction is significant
☒ No, the three-way interaction is not significant

- (d) (3 points) Is it appropriate to test the main effect of Sun light based on the Effect Tests Table?

☐ Yes, because the interaction between Sun light and Type of Soil is not significant.
☒ No, because the interaction between Amount of Fertilizer and Sun light is significant.

6. (35 points) Data from 2022 includes the prices of hot dogs X and beer Y at major league ballparks. You are tasked with determining if the price of a hot dog can be used to predict the price of a beer at these ballparks. Using the provided data, analyze whether there is a significant relationship between the two variables.

Below is a partial output from JMP. Use this output to answer the questions that follow.

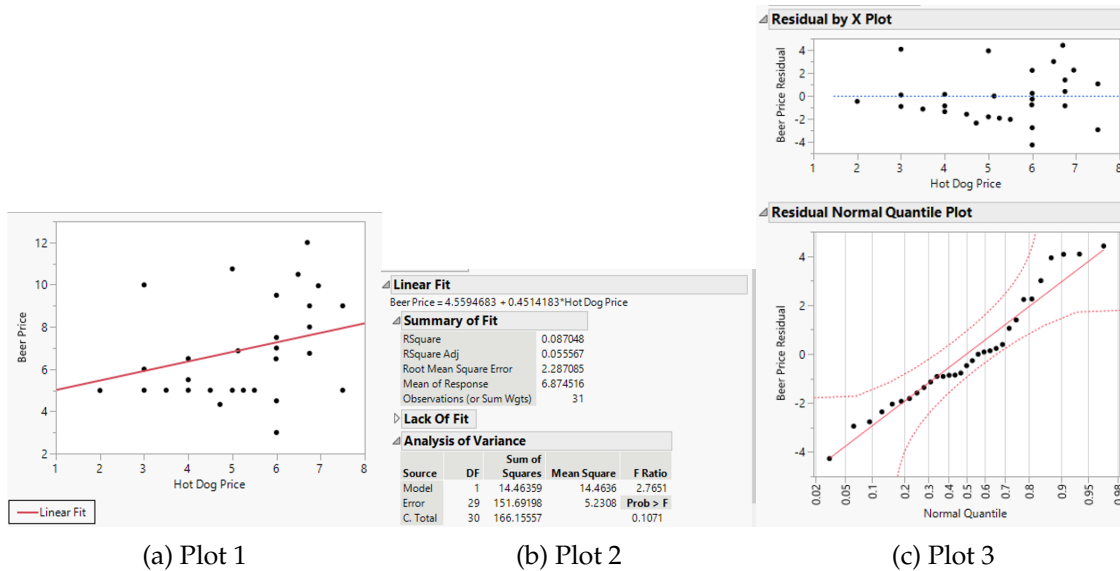


Figure 7: Comparison of three plots related to the hot dog and beer price analysis.

- (a) (4 points) Which of the following is the dependent variable?

☐ Hot Dog Price

☒ Beer Price

- (b) (5 points) State the estimated regression equation (model) that relates the two variables?

Answer = $\hat{\text{Beer Price}} = 4.5594683 + 0.4514183 \times \text{Hot Dog Price}$

- (c) (4 points) What is the expected increase in the beer price for every 1-unit increase in the hot dog price?

Answer = 0.4514

- (d) (9 points) Is there significant evidence of a linear relationship between hot dog price and beer price? Carry out an appropriate test using $\alpha = 0.05$.

(I) Select the appropriate hypothesis.

☐ $H_0 : \beta_1 = 0$
 $H_1 : \beta_1 < 0$

☒ $H_0 : \beta_1 = 0$
 $H_1 : \beta_1 \neq 0$

☐ $H_0 : \beta_0 = 0$
 $H_1 : \beta_0 > 0$

(II) Provide the p-value associated with this test from the ANOVA Table.

P-value =

0.1071

(III) Based on the appropriate p-value reported in the previous question, is there a significant linear relationship between hot dog price and beer price?

- ☐ Yes, there is a significant linear relationship between hot dog price and beer price.
- ☒ No, there is no significant linear relationship between hot dog price and beer price.

- (e) (5 points) What percentage of the variance in beer price is accounted for by hot dog price in the regression model, rounded to two decimal places?

Answer =

$r^2 \times 100\% = 8.7\%$

- (f) (4 points) Do the residuals appear to be approximately normally distributed based on the Residual Normal Quantile Plot?

- ☒ Yes, the points closely align with the diagonal line within the confidence band, indicating a good fit to the normal distribution.
- ☐ No, due to the presence of outliers in the upper and lower tails.

- (g) (4 points) Are there any issues apparent in the Residuals vs. X Plot?

- ☐ No, the residuals are randomly scattered with a consistent spread, indicating constant variance.
- ☒ Yes, the residuals show an increasing spread, suggesting a violation of the assumption of constant variance.